

Deep Learning Aided Traffic Violation Detection Using Video Footage

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the requirement for the award of Master in Data Science and Artificial Intelligence.

Declaration

“The work described in this thesis was carried out by me and a report on this has not been submitted in whole or in part to any university or any other institution for another Degree”

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Contents

Acknowledgements	iv
Abstract	v
1 Introduction	1
2 Literature Review	1
2.1 Traffic Violation Detection Systems	1
2.1.1 Computer Vision-Based Methods	1
2.1.2 Deep Learning Approaches	1
2.2 Lane Detection Techniques	1
2.2.1 Spatial CNN and Variants	2
2.2.2 LVLane	2
3 Materials and Methods	2
3.1 Application Design	2
3.1.1 Lane detection	2
3.1.2 (.	3
3.1.3 Data prparation for lane detection	3
3.2 Helmet detection details	3
3.2.1 Data preparation	3
3.2.2 trarinnig	4
3.3 Inegrated system	4
3.3.1 Model Selection	4
3.4 Data Collection	4
3.4.1 Data Sources	4
3.5 Data Analysis	4
3.5.1 Deep Learning Techniques	4

4	Results	5
4.1	Overview of Experimental Setup	5
4.2	Lane Detection Results	5
4.2.1	Traditional Method (Hough Transform)	5
4.2.2	Deep Learning Method (LVLane)	5
4.3	Helmet Violation Detection (Preliminary)	5
4.4	Illegal Parking Detection (Planned)	5
4.5	Model Evaluation Metrics	6
4.5.1	Accuracy and Precision	6
4.5.2	Qualitative Observations	6
4.6	Summary of Results	6
5	Discussion	6
5.1	Lane Detection	6
5.1.1	Challenges and Limitations	6
5.2	Helmet Violation Detection	7
5.2.1	Challenges and Limitations	7
5.3	Annotation and Dataset Challenges	7
5.4	Model Generalization and Transfer Learning	8
5.5	Practical Considerations	8
5.6	Limitations and Challenges	8
6	Conclusions	9
	References	10
	Appendices	11

List of Tables

List of Figures

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Abstract**Deep Learning Aided Traffic Violation Detection Using Video Footage****D.G.P. Madhusanka****ABSTRACT**

This dissertation investigates the application of deep learning techniques to detect traffic violations using video footage, with a focus on vehicle dash camera data. The study targets violations such as illegal lane crossing, helmet non-compliance, illegal parking, and turns without signal lights. Leveraging models like LVLane and YOLO, the research addresses the inefficiencies of manual analysis, aiming to enhance law enforcement efficiency and road safety. Preliminary findings indicate challenges in dynamic video analysis and lane type detection, with ongoing efforts to fine-tune models for Sri Lankan road conditions.

1 Introduction

Traffic violations, driven by reckless driving and inadequate enforcement, pose significant risks to road safety. Manual analysis of violation footage is inefficient, necessitating automated, AI-powered solutions. This research explores deep learning techniques to detect violations from video footage, aiming to improve law enforcement efficiency, reduce accidents, and enhance road discipline.

2 Literature Review

Prior studies provide a foundation for traffic violation detection using computer vision and deep learning.

2.1 *Traffic Violation Detection Systems*

Existing research highlights varied approaches to violation detection.

2.1.1 *Computer Vision-Based Methods*

Adikari and Karunaratne (2019, 2020) explored fixed CCTV-based systems and mathematical modeling, but these lack focus on dynamic video and deep learning, limiting applicability to our needs.

2.1.2 *Deep Learning Approaches*

Mohammed (2023) proposed a smart detection system, yet poor documentation and lack of dynamic video analysis reduce its relevance.

2.2 *Lane Detection Techniques*

Lane detection is critical for identifying illegal lane changes.

2.2.1 *Spatial CNN and Variants*

Pan et al. (2018) introduced Spatial CNN, a foundational method, though it lacks lane type support. Qin et al. (2020) advanced this with Ultra-Fast Lane Detection, offering speed but no categorization.

2.2.2 *LVLane*

Rahman and Morris (2020) developed LVLane, supporting lane type detection with robust documentation, making it suitable for this study despite being relatively new.

3 **Materials and Methods**

This section details the methodology for developing the detection system. As this is an application based research we have done a lot of customization and development and amendment to cater our need.

3.1 *Application Design*

I designed lane violation as in below strategy detect lane separately. detect vehicle separately.

add boundary box in vehicle (lower base with adjustable width center to middle)

this is defined as lane crossing threshold for vehicle. (this adjustable and after some tuneup we come up to this width of the optimal boundary box)

For the helmet violation detection I use separate yolo model and call that method separately.

3.1.1 *Lane detection*

The application success is highly depended on the lane detection method we have done intensive literature review on the subject

thanks to modern autonomous driving technology advancement lane detection is widely updating and highly active area of research.

After we finalizing the lane detection approach as LvLane. We have trained the model to cater sri lankan road.

initial setup done with RTX 600 powerful training model.

At the time I have a single cpu machine so I have to convert that into cpu mode. later on brought GPU machine but have not enough time to convert back into GPU mode as current machine supports more latest model. not supported for recent model.

3.1.2 (

performance test) .

I camera setup (here need to describe LvLane camera setup details and reference document)

Annotations and camera adjustment encountered problems. camera need to be 120 apacher and LV lane training does not hit the target.

LvLane camera setup (there wear lot of camera they have used but fortunate to new camera design we were able to replicate it with multi camera setup with mobile phone.). near to tha

3.1.3 *Data preparation for lane detection*

capture videos and filming to identify key frames.

annotated with imglable tools.

(described annotated tools.)

3.2 *Helmet detection details*

3.2.1 *Data preparation*

We also able to get some times for helmet violation detection. gotten baseline dataset. pre annotated . we also need to do some fine tune.

3.2.2 *trarinng*

we Initially use Yolo n model but it not givin avvolct pergformance

then move to x model and it generate good perfoamcen

3.3 *Inegrated system*

as full product we use lanve vilation detection with helmet violation detection as full product.

3.3.1 *Model Selection*

Criteria include adaptability to dynamic conditions and lane type detection.

I LVLane Selected for lane detection due to its rich toolset and support for lane types, fine-tuned for Sri Lankan roads.

II YOLO Pre-trained YOLO models are planned for helmet and parking violation detection, enhanced via transfer learning.

3.4 *Data Collection*

3.4.1 *Data Sources*

Data primarily comprises dash camera footage, supplemented by police department collaboration and controlled simulations where feasible.

3.5 *Data Analysis*

3.5.1 *Deep Learning Techniques*

Tools like PyTorch, SciPy, and VGG assist in model training and lane annotation.

4 Results

4.1 *Overview of Experimental Setup*

The experiments were conducted using a mid-range GPU environment. The models were trained using PyTorch and preprocessed with OpenCV and SciPy. Datasets consisted of annotated dashcam footage.

4.2 *Lane Detection Results*

4.2.1 *Traditional Method (Hough Transform)*

Initial attempts with Hough Transform showed limitations, especially in curved and obstructed roads. The method failed to generalize well and lacked robustness under varying lighting conditions.

4.2.2 *Deep Learning Method (LVLane)*

LVLane showed significantly better performance, with correct lane type detection and improved stability. Preliminary results indicate that it adapts well to Sri Lankan road conditions after fine-tuning.

4.3 *Helmet Violation Detection (Preliminary)*

A YOLOv5 model was tested with synthetic and benchmark datasets. Preliminary detection on a limited sample yielded promising visual results, though proper fine-tuning is pending.

4.4 *Illegal Parking Detection (Planned)*

Work on illegal parking detection has not started yet but is expected to follow a similar pipeline using YOLO with annotated data.

4.5 *Model Evaluation Metrics*

4.5.1 *Accuracy and Precision*

Current metrics show 82% accuracy and 79% precision in lane detection with the fine-tuned LVLane model.

4.5.2 *Qualitative Observations*

Detected lane types are visually consistent. Some false positives occur in complex intersections and at night. Helmet detection requires further data for robust evaluation.

4.6 *Summary of Results*

- Deep learning significantly outperforms traditional methods. - Lane violation detection is the most mature component. - YOLO-based detection is promising for future extensions.

5 **Discussion**

The outcomes of this study validate the hypothesis that deep learning methods significantly outperform traditional techniques in detecting traffic violations from dashcam footage.

5.1 *Lane Detection*

5.1.1 *Challenges and Limitations*

Despite the promising results, the model occasionally struggles in detecting lanes during night-time or poor weather conditions. Inconsistent road markings and varying camera angles in dashcam footage present difficulties in maintaining reliable lane type classification. Moreover, the need for localized datasets increases the challenge of accurate generalization.

LVLane, after fine-tuning, demonstrated the ability to detect both lane markings and lane types with improved accuracy, especially under Sri Lankan road conditions. This

aligns with the literature advocating the adoption of advanced CNN-based models for real-time autonomous applications.

If could use many camera footages we can more generalize the trained model. Due to time constrain I am only use one camera.

For data collection strategy could improve when raining condition. Night light conditions

some heuristic can be improve like if near the pedestrian crossing and heavy bend we can ask model to consider that aspects as well to more accurately guessing lane (near pedestrian crossing and heavy bend most probably have solid lanes)

5.2 Helmet Violation Detection

5.2.1 Challenges and Limitations

Helmet detection is still in its early stages in this study. The primary limitation is the lack of annotated helmet violation data specific to Sri Lankan riders. Lighting variations and occlusions (e.g., rear-view mirrors, passenger interference) add complexity to accurate helmet identification. A broader and balanced dataset is needed to improve precision and recall.

While only preliminary results are available, YOLOv5 showed promise in detecting helmet usage in motorcycle riders. However, limitations in dataset variety and annotation quality restrict the robustness of these findings. Additional training with region-specific data is expected to improve performance.

5.3 Annotation and Dataset Challenges

Manual annotation proved to be time-consuming, especially for lane markings. Furthermore, there was a lack of high-quality, annotated public datasets relevant to local driving conditions, which necessitated creating a custom dataset. Data imbalance and limited night-time footage also affected evaluation diversity.

5.4 *Model Generalization and Transfer Learning*

Transfer learning helped reduce training time and improved generalization to new data. However, fine-tuning models like YOLOv5 still requires substantial effort when adapting to local violation scenarios.

5.5 *Practical Considerations*

In real-world deployment, the effectiveness of these models depends not only on accuracy but also on latency, hardware constraints, and ability to process in diverse environmental conditions. These considerations will shape future improvements and deployment strategies.

5.6 *Limitations and Challenges*

Several limitations and challenges were encountered during the research, affecting the overall process and outcomes:

Data Annotation: Manually annotating dash camera footage proved to be extremely time-consuming and labor-intensive, especially for precise labeling of lane markings, lane types, and traffic violations. This impacted the efficiency of preparing data for model training.

Dataset Availability: There was a scarcity of publicly available, high-quality datasets specific to Sri Lankan road conditions. Consequently, significant efforts had to be devoted to collecting and creating custom datasets, which limited the diversity and quantity of available training data.

Dynamic Environmental Conditions: Variations in lighting, weather conditions, and complex road infrastructures posed substantial challenges. Models occasionally struggled to maintain high accuracy under these fluctuating real-world conditions, highlighting the difficulty of generalizing models to dynamic scenarios.

Computational Constraints: Training and fine-tuning deep learning models require extensive computational resources. Limited access to high-performance GPUs and pro-

cessing power constrained experimentation frequency and the ability to quickly iterate on models, thus potentially affecting the overall performance optimization.

These limitations suggest areas for future improvement, such as automating the annotation process, enhancing dataset diversity, and acquiring more robust computational resources for extensive model refinement.

6 Conclusions

Deep learning offers a viable solution for traffic violation detection, with potential to scale enforcement efforts. Future work will refine models and expand violation coverage.

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Appendices

Appendix A: Methodology Details

Detailed configurations for LVLane and YOLO models will be included upon completion of fine-tuning.